

Eilenberg-Moore Categories of Markov Monads

Rob Cornish

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June 9, 2025

Outline

A (mostly finished) work in progress with two main parts: one theoretical, one application.

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Application

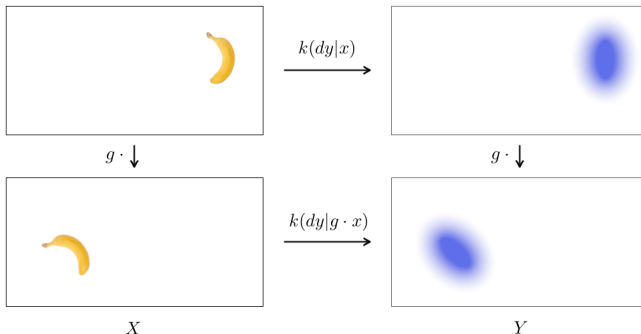
We use this result to give a conceptual treatment of some representation theorems from the AI literature by Bloem-Reddy and Teh [2020]

Motivation (previous work)

Stochastic Neural Network Symmetrisation in Markov Categories

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Background

Markov categories

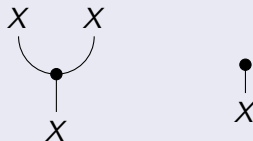
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Definition (Cho and Jacobs [2019], Fritz [2020])

A Markov category $\mathbf{C}$ is a symmetric monoidal category in which each object X is equipped with a suitably compatible **commutative comonoid** ($\mathbf{copy}_X, \mathbf{del}_X$), denoted in string diagrams by

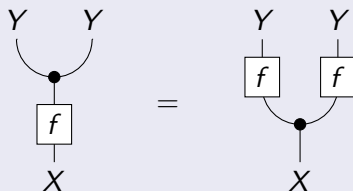


Determinism

Markov categories model stochastic processes, but as an important degenerate case, we have the following:

Definition (Fritz [2020])

A morphism $f : X \rightarrow Y$ is **deterministic** if



Can show that the deterministic morphisms form a **cartesian monoidal** subcategory $\mathbf{C}_{\text{det}} \subseteq \mathbf{C}$

Theory: Markov monads

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For T -algebras to form a Markov category, it is clear T must respect the Markov structure in some way

This means:

- The symmetric monoidal structure
- The commutative comonoids (copy maps)

Markov functors (idea)

Let $F : \mathbf{C} \rightarrow \mathbf{D}$ be functor between symmetric monoidal categories, and let

$$\Delta_{X,Y} : F(X \otimes Y) \rightarrow FX \otimes FY$$

a **colax monoidal** structure on F (with $\delta : FI \rightarrow I$ uniquely given)

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Recall this **transport comonoids**: if $c : X \rightarrow X \otimes X$ is a comultiplication in $\mathbf{C}$, then

$$FX \xrightarrow{Fc} F(X \otimes X) \xrightarrow{\Delta} FX \otimes FX$$

is one in $\mathbf{D}$ also

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Say that (F, Δ) is **Markov** if this sends **copy** _{X} specifically to **copy** _{FX}

Markov functors (formally)

Definition

A **Markov functor** $(F, \Delta) : \mathbf{C} \rightarrow \mathbf{D}$ is a colax symmetric monoidal functor such that the following diagram commutes for all objects X in $\mathbf{C}$:

$$\begin{array}{ccc} & FX & \\ F(\text{copy}_X) \swarrow & & \searrow \text{copy}_{FX} \\ F(X \otimes X) & \xrightarrow{\Delta_{X,X}} & FX \otimes FX \end{array}$$

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This slightly generalises the definition given by Fritz [2020]

Characterisation

Theorem

Suppose $(F, \Delta) : \mathbf{C} \rightarrow \mathbf{D}$ is Markov functor. Then:

- 1 F restricts to a functor $F_{\text{det}} : \mathbf{C}_{\text{det}} \rightarrow \mathbf{D}_{\text{det}}$.
- 2 Each structure map $\Delta_{X,Y}$ is given by universal property of the product in $\mathbf{D}_{\text{det}}$ as follows:

$$\begin{array}{ccccc} & & F(X \otimes Y) & & \\ & \swarrow F\pi_X & \downarrow \Delta_{X,Y} & \searrow F\pi_Y & \\ FX & \xleftarrow{\pi_{FX}} & FX \otimes FY & \xrightarrow{\pi_{FY}} & FY \end{array}$$

where we denote projection maps by π

Idea of proof

Three key steps:

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- 1 Show that $\Delta_{X,Y}$ is deterministic (similar idea to Fong and Spivak [2020])
- 2 Show that F restricts to F_{det} (diagram chase)
- 3 Use the fact that functors between cartesian categories are uniquely colax monoidal, with $\Delta_{X,Y} = \langle F\pi_X, F\pi_Y \rangle$

Characterisation (converse)

Theorem

Let $F : \mathbf{C} \rightarrow \mathbf{D}$ be any functor that restricts to $F_{\text{det}} : \mathbf{C}_{\text{det}} \rightarrow \mathbf{D}_{\text{det}}$, and let $\Delta_{X,Y}$ be its unique colax structure. Then (F, Δ) is Markov whenever the following naturality condition holds for all f and g in $\mathbf{C}$:

$$\begin{array}{ccc} F(X \otimes Y) & \xrightarrow{F(f \otimes g)} & F(X' \otimes Y') \\ \downarrow \Delta_{X,Y} & & \downarrow \Delta_{X',Y'} \\ FX \otimes FY & \xrightarrow{Ff \otimes Fg} & FX' \otimes FY' \end{array}$$

(as opposed to just f and g in $\mathbf{C}_{\text{det}}$).

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(as opposed to just f and g in $\mathbf{C}_{\text{det}}$).

Here Δ can only be one thing, but I don't know if naturality is guaranteed!

Markov monads

Definition

A **Markov monad** is a monad in the 2-category of Markov categories, Markov functors, and colax monoidal natural transformations.

The 2-morphisms here can be equivalently characterised as follows:

Proposition

Let $(F, \Delta), (G, \Delta') : \mathbf{C} \rightarrow \mathbf{D}$ be Markov functors. A natural transformation $F \Rightarrow G$ is colax monoidal if and only if has deterministic components.

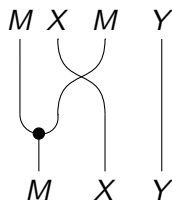
Example

Given a monoid M in $\mathbf{C}_{\text{det}}$, the **writer monad** $M \otimes (-)$ is a Markov monad on $\mathbf{C}$

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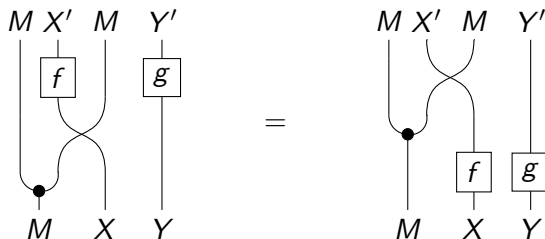
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The monoidal structure $\Delta_{X,Y}$ is given by



Example (continued)

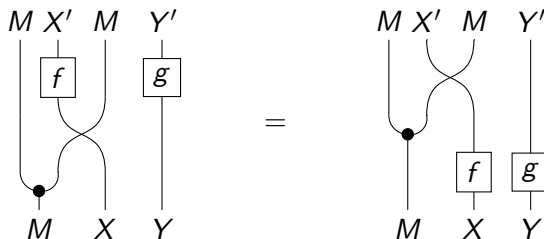
The naturality condition for Δ becomes:



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More examples can be obtained via **distributive laws** $T_{\text{det}}P \Rightarrow PT_{\text{det}}$, where T_{det} and P are monads on a cartesian category

Symmetric monoidal Eilenberg-Moore categories

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The following folkloric (?) result gets us part of the way there:

Proposition

If T is a colax symmetric monoidal monad, then its Eilenberg-Moore category $\mathbf{C}^T$ is also symmetric monoidal, where its monoidal product $\otimes_T$ is given on objects by

$$(X, \alpha) \otimes_T (Y, \beta) := (X \otimes Y, (\alpha \otimes \beta) \circ \Delta_{X,Y})$$

and on morphisms simply by $f \otimes_T g := f \otimes g$.

Lifting copy maps

Therefore, the question is whether $\mathbf{C}^T$ has copy maps

$$\mathbf{copy}_{(X, \alpha)} : (X, \alpha) \rightarrow (X, \alpha) \otimes (X, \alpha)$$

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Hence the only candidate is simply $\mathbf{copy}_X$

However, often $\mathbf{copy}_X$ is not an algebra homomorphism!

A problem

E.g. for the writer monad $M \otimes (-)$, the following must commute:

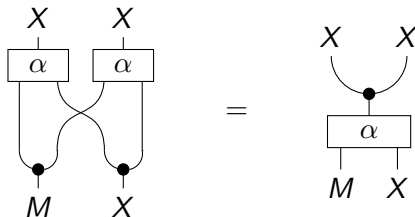
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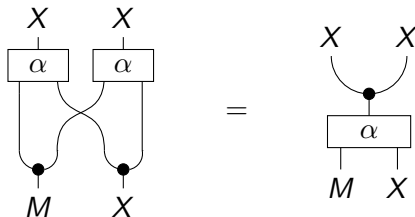


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This says precisely that α is **deterministic**. However, often not true!

Workaround

We can work around this issue by only considering algebras (X, α) for which α is **deterministic**

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We then have the following:

Theorem

If T is a Markov monad on $\mathbf{C}$, then $\mathbf{C}_{\text{det}}^T$ is a Markov category, with

- $(X, \alpha) \otimes (Y, \beta) := (X \otimes Y, (\alpha \otimes \beta) \circ \Delta_{X,Y})$ as before
- Copy and deletion maps inherited from $\mathbf{C}$ (e.g. $\mathbf{copy}_{(X, \alpha)} := \mathbf{copy}_X$)

Application

A representation theorem for neural networks

Theorem (Bloem-Reddy and Teh [2020] (distributional version))

Let G be a compact group acting measurably on standard Borel spaces X and Y . Suppose $\mathbf{X}$ and $\mathbf{Y}$ are random variables taking values in X and Y respectively, and that moreover $g \cdot \mathbf{X} \stackrel{d}{=} \mathbf{X}$ for all $g \in G$. Likewise, let $\mathbf{U} \sim \text{Uniform}(0, 1)$ be independent of $\mathbf{X}$. Then we have

$$(g \cdot \mathbf{X}, g \cdot \mathbf{Y}) \stackrel{d}{=} (\mathbf{X}, \mathbf{Y}) \quad \text{for all } g \in G \quad (1)$$

if and only if we can write

$$(\mathbf{X}, \mathbf{Y}) \stackrel{d}{=} (\mathbf{X}, f(\mathbf{X}, \mathbf{U})). \quad (2)$$

for some measurable function $f : X \times [0, 1] \rightarrow Y$ satisfying

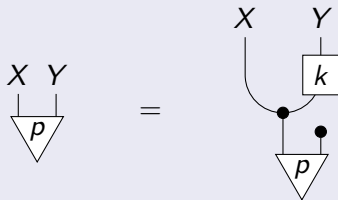
$$f(g \cdot x, \mathbf{U}) \stackrel{d}{=} g \cdot f(x, \mathbf{U}) \quad \text{for all } g \in G \text{ and } x \in X. \quad (3)$$

Randomness pushback

To state our version of this result, we need the following definitions:

Definition (Cho and Jacobs [2019], Fritz [2020])

A Markov category $\mathbf{C}$ has **conditional distributions** if every morphism $p : I \rightarrow X \otimes Y$ in $\mathbf{C}$ can be written in the form



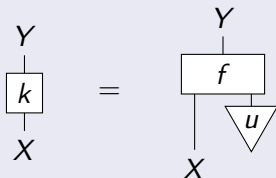
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The diagram shows an equality between two morphisms from X to Y . On the left, a box labeled k has an input from X and an output to Y . On the right, a box labeled f has an input from X and another input from a triangle labeled u (representing the identity on the monoidal unit I). The output of f is Y . The two diagrams are separated by an equals sign.

for some $u : I \rightarrow U$ and some **deterministic** $f : X \otimes U \rightarrow Y$ in $\mathbf{C}$

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Theorem

Let $\mathbf{C}$ be a Markov category with conditional distributions and randomness pushback. For any compact group G in $\mathbf{C}$, the Markov category $\mathbf{C}^G$ has conditional distributions and randomness pushback also.

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As a special case, **BorelStoch** ^{G} has conditional distributions and randomness pushback

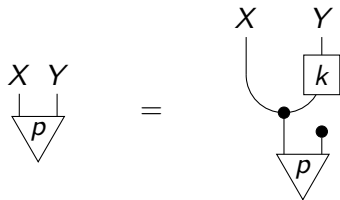
Proof idea for conditionals

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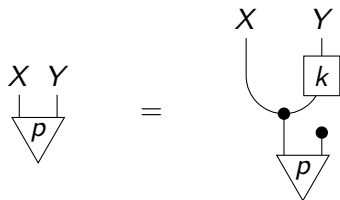
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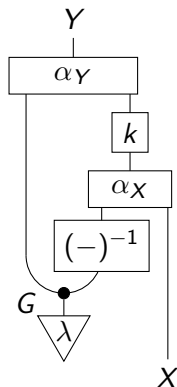
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Can show that k is $(\pi_X \circ p)$ -a.s. equivariant wrt. α_X and α_Y

Proof idea for conditionals (continued)

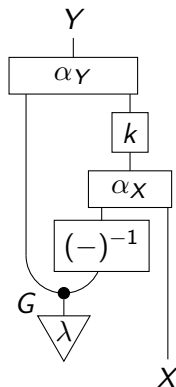
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where $\lambda : I \rightarrow G$ denotes the Haar measure, and $(-)^{-1} : G \rightarrow G$ the group inversion

Proof idea for conditionals (continued)

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Can show this is *pointwise* equivariant, and also $(\pi_X \circ p)$ -a.s. equal to k , hence a conditional of p

Implications for Bloem-Reddy and Teh [2020]

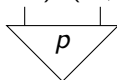
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Then $(g \cdot \mathbf{X}, g \cdot \mathbf{Y}) \stackrel{d}{=} (\mathbf{X}, \mathbf{Y})$ iff p is a morphism in **BorelStoch** ^{G} of form

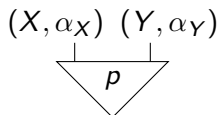
$$(X, \alpha_X) \quad (Y, \alpha_Y)$$



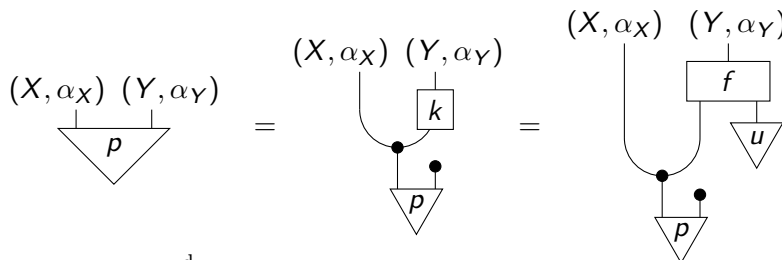
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Since **BorelStoch**^G has cond. distrs. and randomness pushback, we obtain



which says $(\mathbf{X}, \mathbf{Y}) \stackrel{d}{=} (\mathbf{X}, f(\mathbf{X}, \mathbf{U}))$ where f is equivariant

Thank you!

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I'm building a new research group at **Nanyang Technological University**



NTU is a great place to do research! (Top 5 in Asia, top 30 globally)

I have **fully funded PhD scholarships** and **postdoc positions** available in:

- AI/machine learning (uncertainty quantification, equivariance, etc.)
- Its intersection with category theory
- Its intersection with proof assistants (e.g. Lean)

References I

- Benjamin Bloem-Reddy and Yee Whye Teh. Probabilistic symmetries and invariant neural networks. *Journal of Machine Learning Research*, 21(90):1–61, 2020. URL <http://jmlr.org/papers/v21/19-322.html>.
- Kenta Cho and Bart Jacobs. Disintegration and bayesian inversion via string diagrams. *Mathematical Structures in Computer Science*, 29(7):938–971, March 2019. ISSN 1469-8072. doi: 10.1017/s0960129518000488. URL <http://dx.doi.org/10.1017/S0960129518000488>.
- Tobias Fritz. A synthetic approach to markov kernels, conditional independence and theorems on sufficient statistics. *Advances in Mathematics*, 370:107239, August 2020. ISSN 0001-8708. doi: 10.1016/j.aim.2020.107239. URL <http://dx.doi.org/10.1016/j.aim.2020.107239>.
- Brendan Fong and David I Spivak. Supplying bells and whistles in symmetric monoidal categories, 2020. URL <https://arxiv.org/abs/1908.02633>.

Appendix

Another perspective on Markov functors

Corollary

A Markov functor $(F, \Delta) : \mathbf{C} \rightarrow \mathbf{D}$ is precisely a **colax extension** of a functor $F_{\text{det}} : \mathbf{C}_{\text{det}} \rightarrow \mathbf{D}_{\text{det}}$

Here by “colax extension” I mean a colax symmetric monoidal functor $(F, \Delta) : \mathbf{C} \rightarrow \mathbf{D}$ such that the following commutes:

$$\begin{array}{ccc} \mathbf{C}_{\text{det}} & \xrightarrow{F_{\text{det}}} & \mathbf{D}_{\text{det}} \\ \downarrow & & \downarrow \\ \mathbf{C} & \xrightarrow{F} & \mathbf{D} \end{array}$$

It automatically follows that Δ is the (unique) colax symmetric monoidal structure of F_{det}

Distributive laws

Many Markov categories $\mathbf{C}$ arise via the following configuration:

- ① We have a symmetric monoidal monad P on a cartesian monoidal $\mathbf{D}$
- ② We obtain $\mathbf{C} := \mathbf{Kl}(P)$

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Then we have the following correspondences:

Distributive laws $T_{\text{det}} P \Rightarrow P T_{\text{det}} \longleftrightarrow$ Extensions of T_{det}

Markov monads $T \longleftrightarrow$ Colax extensions of T_{det}

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- 1 We have a symmetric monoidal monad P on a cartesian monoidal $\mathbf{D}$
- 2 We obtain $\mathbf{C} := \mathbf{Kl}(P)$

Suppose moreover the inclusion $\mathbf{D} \hookrightarrow \mathbf{C}$ gives an isomorphism $\mathbf{D} \cong \mathbf{C}_{\text{det}}$
(i.e. $\mathbf{C}$ is **representable**)

Then we have the following correspondences:

Distributive laws $T_{\text{det}} P \Rightarrow P T_{\text{det}} \longleftrightarrow$ Extensions of T_{det}

Markov monads $T \longleftrightarrow$ Colax extensions of T_{det}

(Without representability, need to be more careful about T_{det} on $\mathbf{D}$ vs. $\mathbf{C}_{\text{det}}$)

Examples

This gives a source of Markov monads:

Procedure

- 1 Choose a symmetric monoidal monad P on a cartesian $\mathbf{D}$
- 2 Choose a monad T_{det} on $\mathbf{D}$ with distributive law $\delta : T_{\text{det}}P \Rightarrow PT_{\text{det}}$
- 3 Extend T_{det} via δ to obtain a monad T on $\mathbf{C} := \mathbf{Kl}(P)$
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Concrete example: the **maybe monad** $I + (-)$ on **BorelStoch**

Alternative strategy

With a distributive law $\delta : T_{\text{det}} P \Rightarrow P T_{\text{det}}$, we could also lift P to a monad $\tilde{P}$ on $\mathbf{C}^{T_{\text{det}}}$

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In fact, can show that naturality condition on Δ implies $\tilde{P}$ is symmetric monoidal (hence $(\mathbf{C}^{T_{\text{det}}})_{\tilde{P}}$ is Markov). I don't know about the converse!

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Can show this is always a morphism in $\mathbf{C}^G$ of the form

$$f : (X, \alpha) \otimes (U, \varepsilon) \rightarrow (Y, \beta)$$